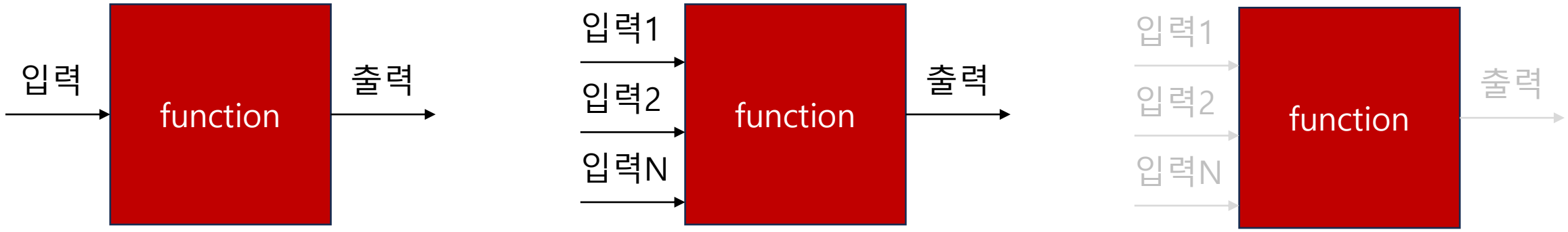
인터럽트 & PWM

마이크로프로세서 종합 설계. 8주차.

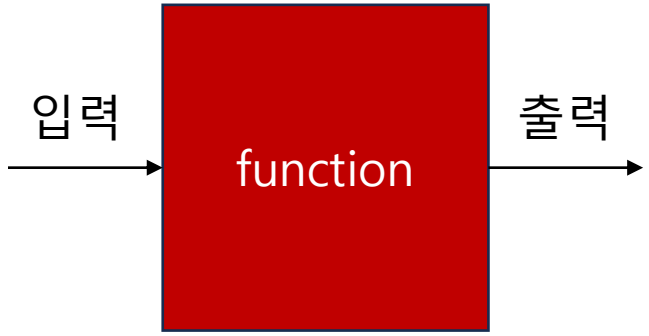
오늘의 목표

1. **C언어 함수란?**
2. **API를 이용한 디지털 입출력(폴링)**
3. **외부인터럽트의 이해 및 인터럽트 실험 (폴링 vs 인터럽트)**
4. **PWM의 이해와 PWM을 이용한 전력 제어 실험**

마이크로프로세서와 C언어 - 함수

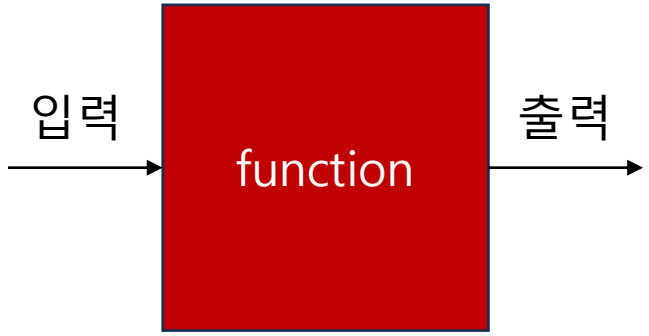


마이크로프로세서와 C언어 - 함수




```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

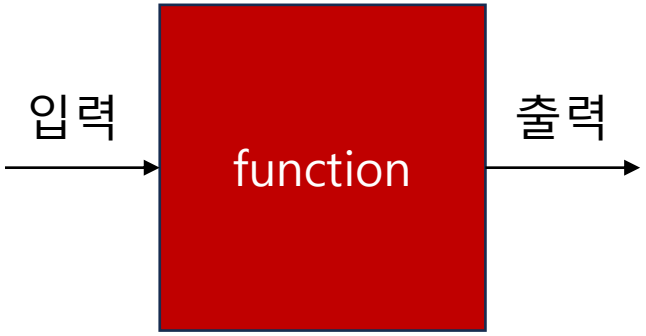
마이크로프로세서와 C언어 - 함수



출력 자료형(return type)

```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

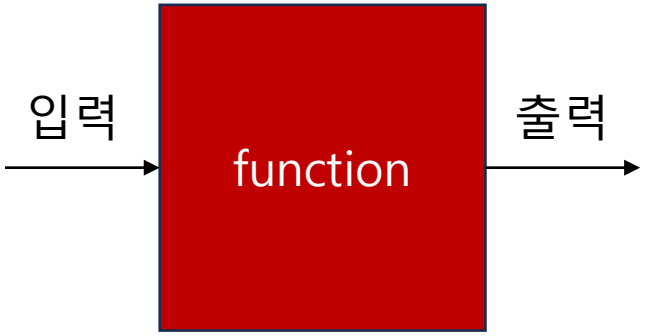
마이크로프로세서와 C언어 - 함수



함수 이름(함수명)

```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

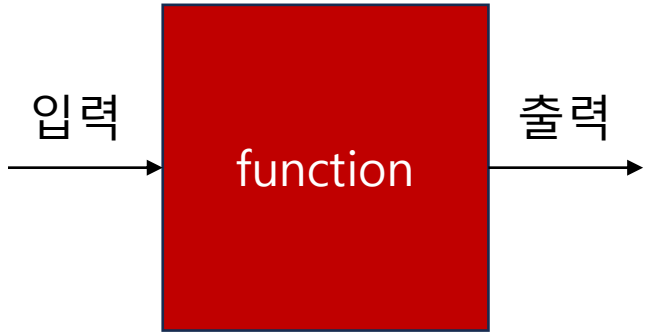
마이크로프로세서와 C언어 - 함수



입력 인자(input argument)

```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

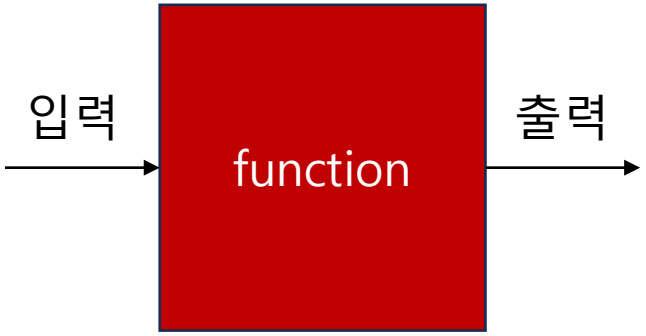
마이크로프로세서와 C언어 - 함수




```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

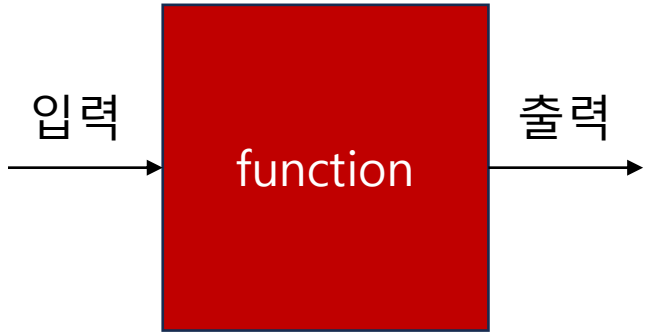
명령문

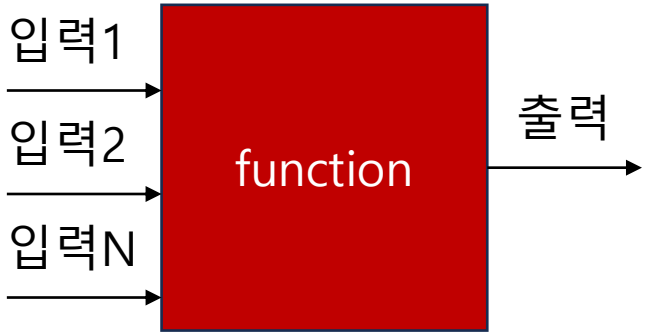
마이크로프로세서와 C언어 - 함수

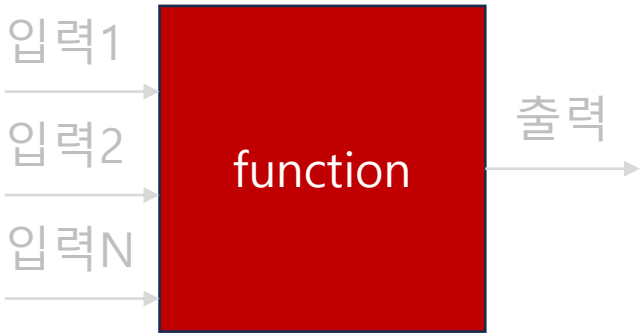



```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

마이크로프로세서와 C언어 - 함수







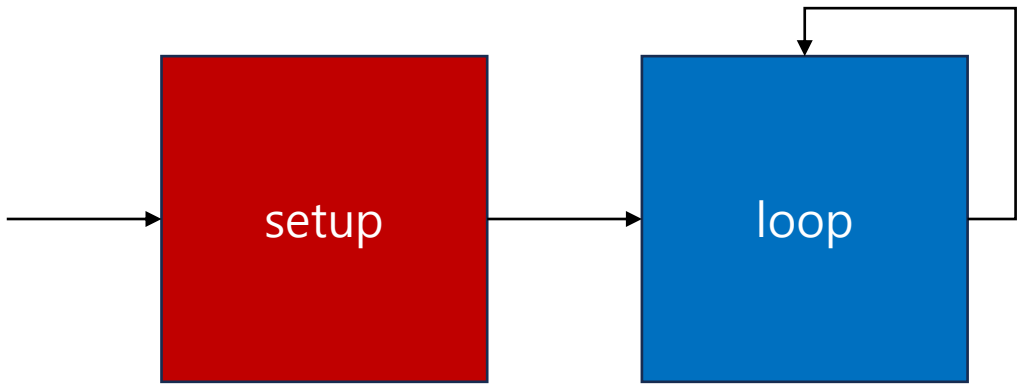

```
int function(int a)
{
    a = a + 10 ;
    return a ;
}
```

```
int function(int a, int b)
{
    int c = a + b ;
    return c ;
}
```

```
void function(void)
{
    //함수 명령
}
```

마이크로프로세서와 C언어 - 함수

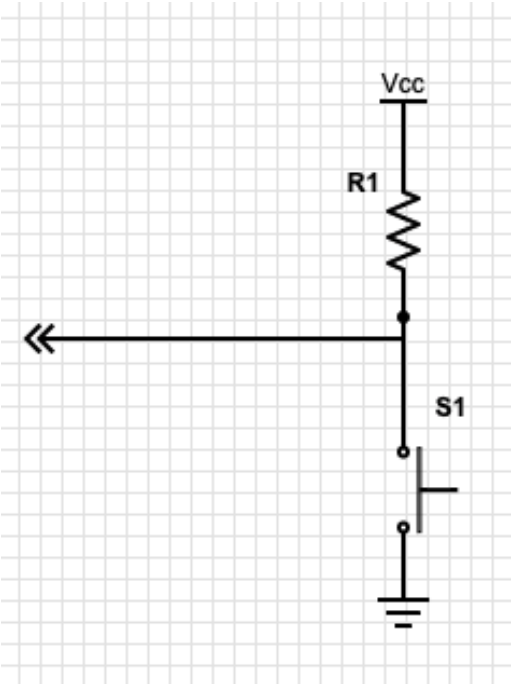
```
void setup()
{
    설정 ;
    초기화 ;
}

void loop()
{
    반복 기능 ;
}
```

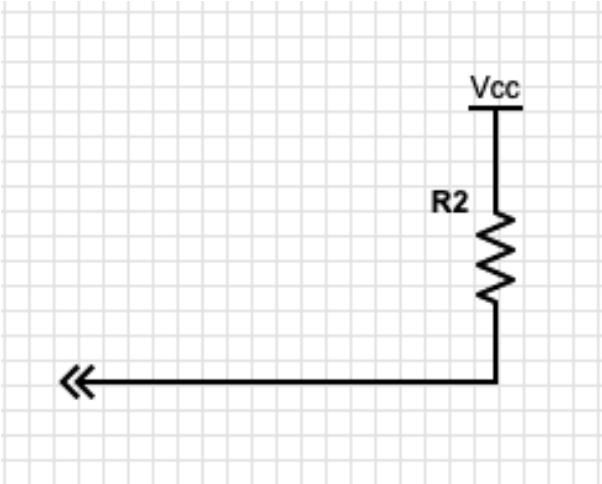


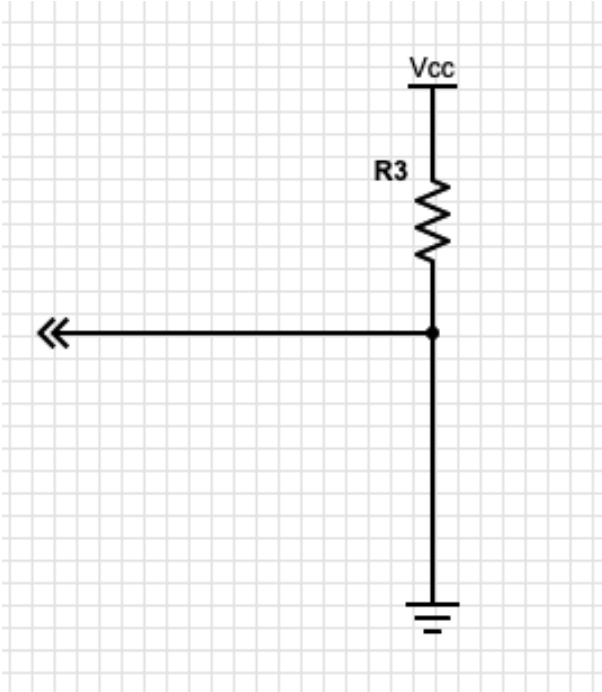
마이크로프로세서와 C언어 - 함수

- **pinMode(8, INPUT) ;**
- **int value = digitalRead(8) ;**
- **pinMode(8, OUTPUT) ;**
- **digitalWrite(8, 1) ;**

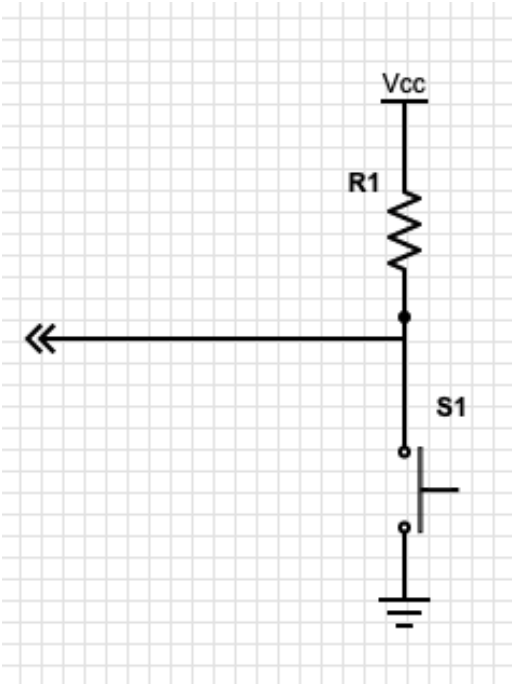
버튼 입력 실험(Digital Input)

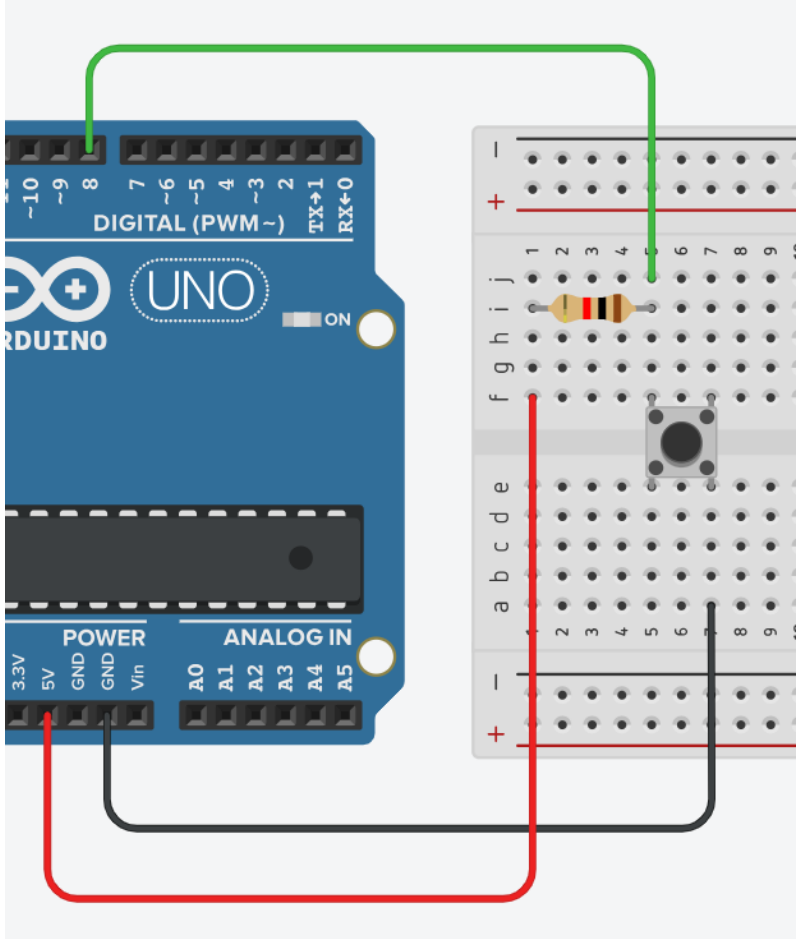




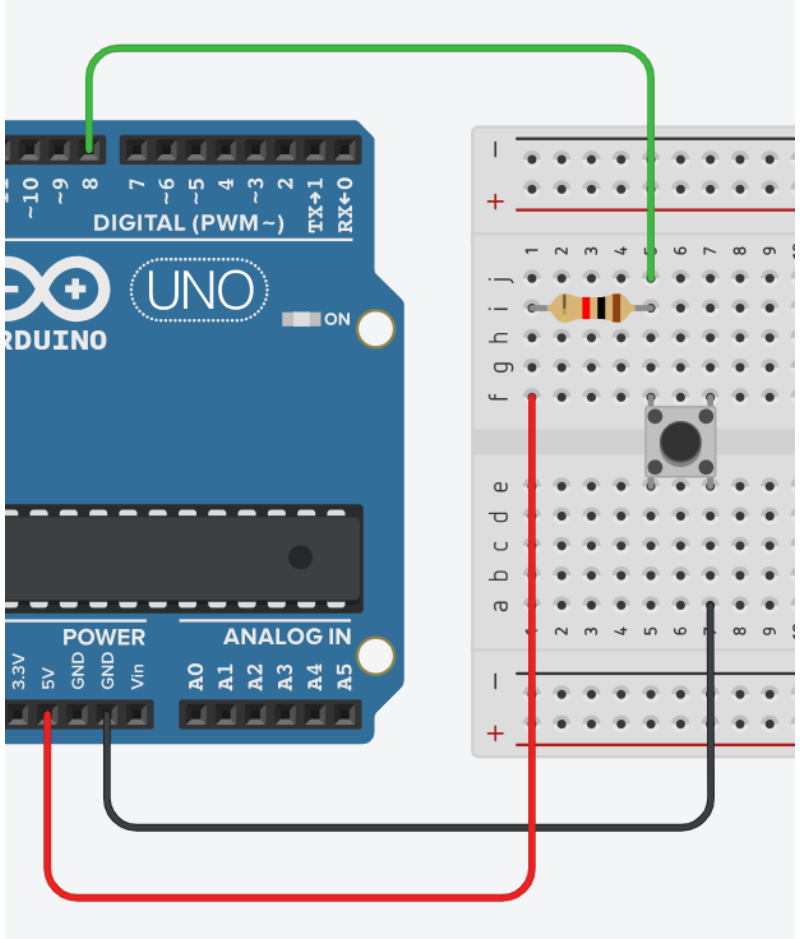


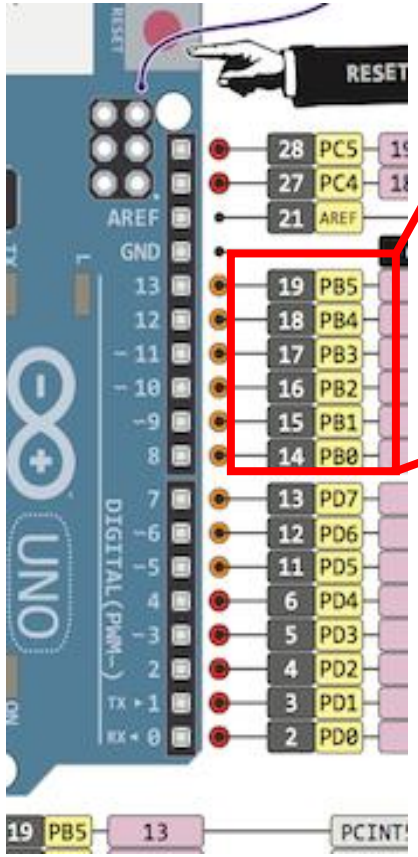
버튼 입력 실험(Digital Input)





버튼 입력 실험(Digital Input)





13.4.2 PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

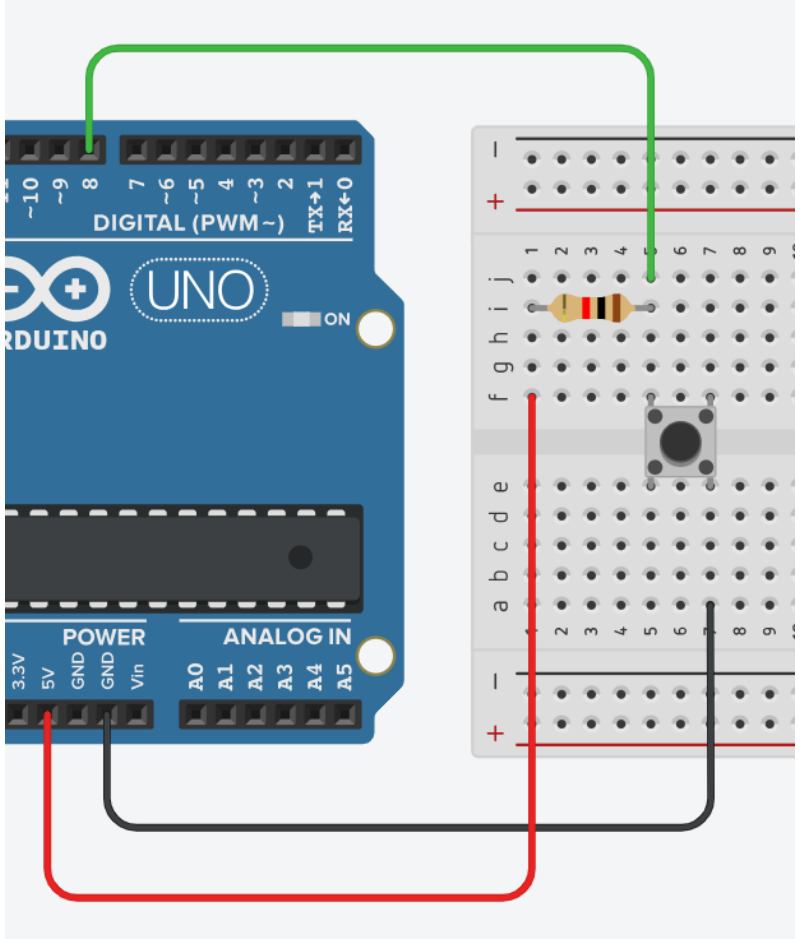
13.4.3 DDRB – The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.4 PINB – The Port B Input Pins Address

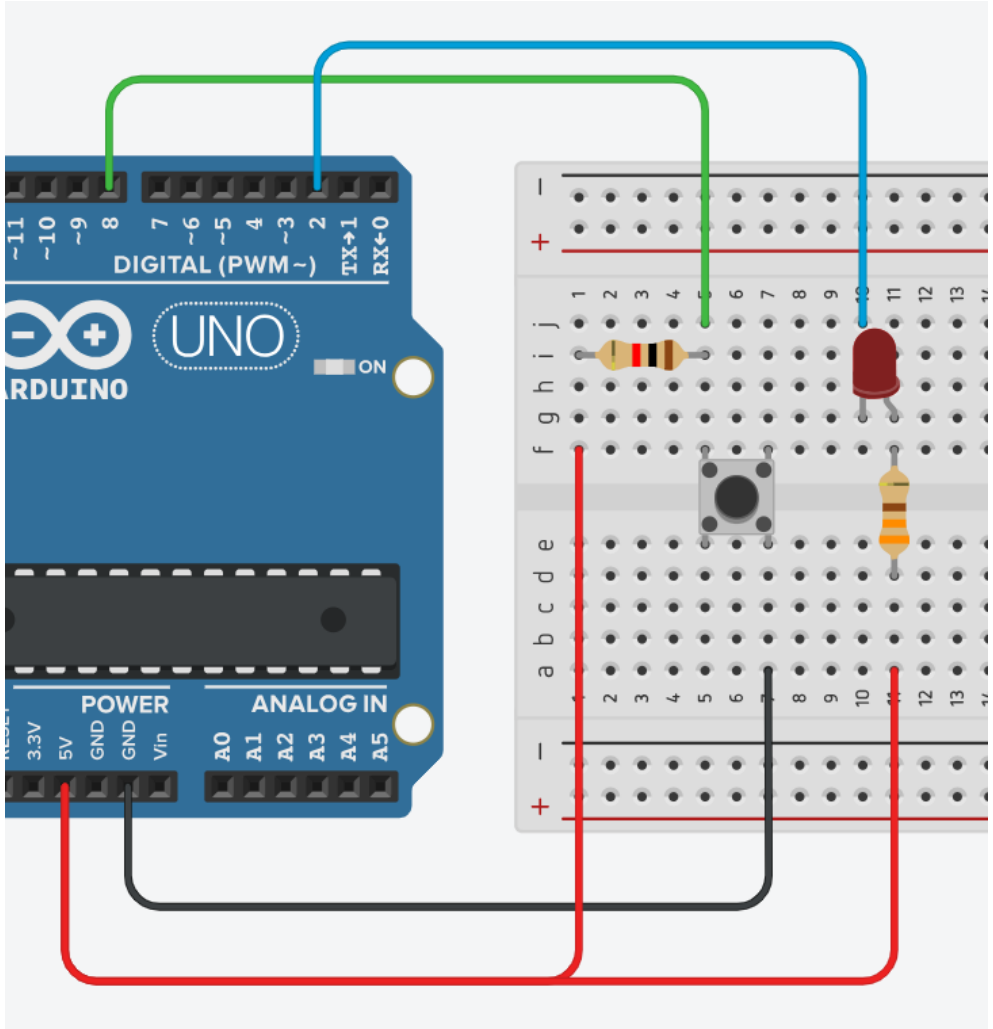
Bit	7	6	5	4	3	2	1	0	
0x03 (0x23)	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	PINB
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

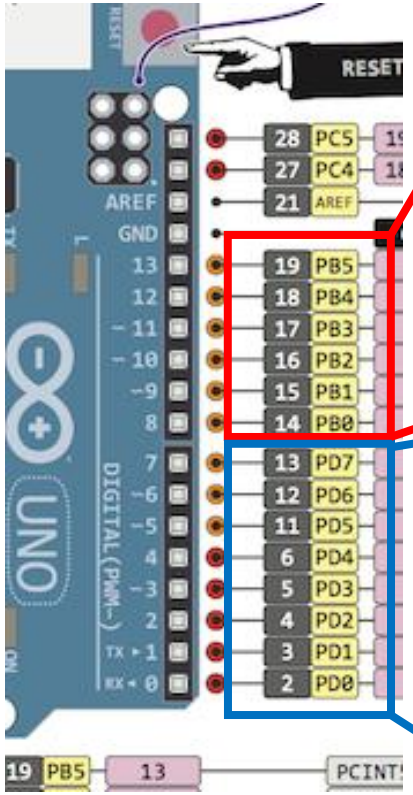
버튼 입력 실험(Digital Input)




```
void setup()
{
  DDRB = B00000000 ;
  Serial.begin(9600) ;
}

void loop()
{
  Serial.println(PINB) ;
}
```

버튼 입력 실험(Digital Input)





13.4.2 PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.3 DDRB – The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.4 PINB – The Port B Input Pins Address

Bit	7	6	5	4	3	2	1	0	
0x03 (0x23)	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	PINB
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

13.4.8 PORTD – The Port D Data Register

Bit	7	6	5	4	3	2	1	0	
0x0B (0x2B)	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0	PORTD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

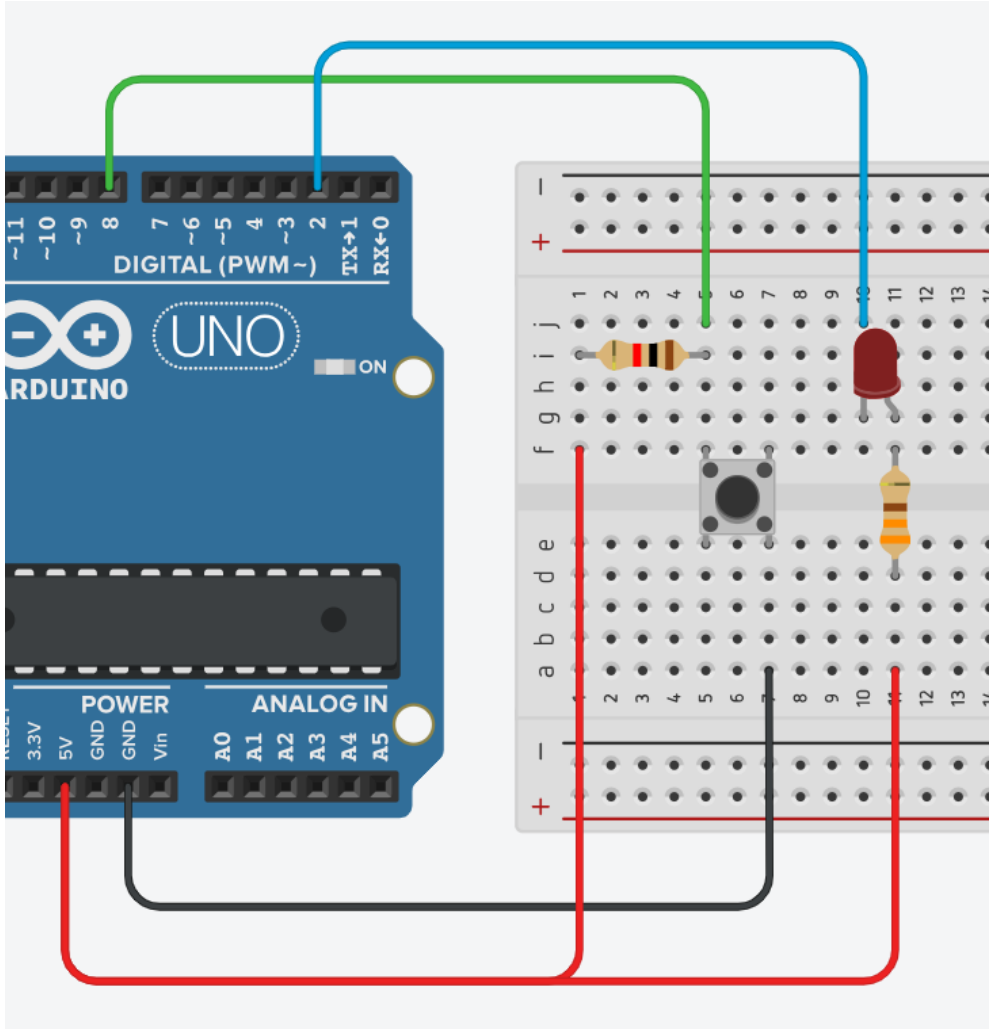
13.4.9 DDRD – The Port D Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.10 PIND – The Port D Input Pins Address

Bit	7	6	5	4	3	2	1	0	
0x09 (0x29)	PIND7	PIND6	PIND5	PIND4	PIND3	PIND2	PIND1	PIND0	PIND
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

버튼 입력 실험(Digital Input)




```
void setup()
{
  DDRB = B00000000 ;
  DDRD = B00000100 ;

  Serial.begin(9600) ;
}

void loop()
{
  Serial.println(PINB) ;

  if( PINB == B00000001 )
  {
    //버튼이 눌리지 않음
    PORTD = B0000100 ;          //LED 꺼짐
  }
  else
  {
    //버튼이 눌림
    PORTD = B0000000 ;          //LED 켜짐
  }
}
```

디지털 입출력 관련 API

- pinMode
- digitalRead
- digitalWrite

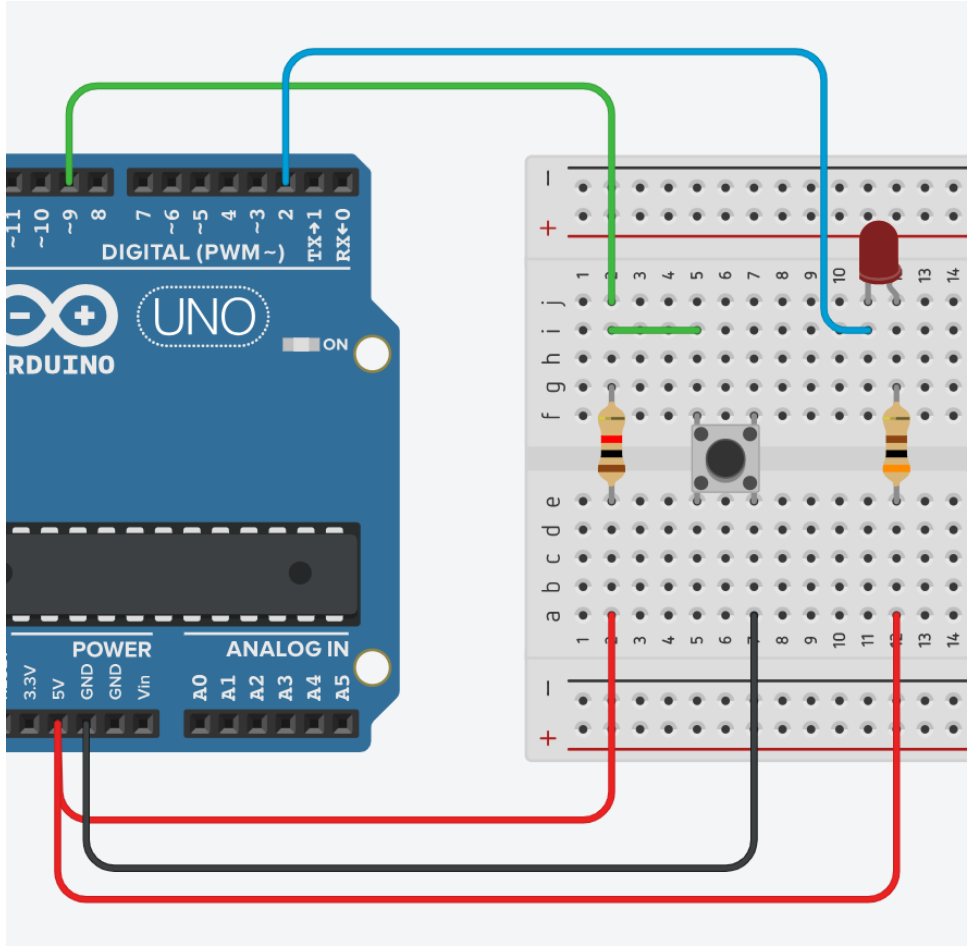
디지털 입출력 관련 API

- 디지털 입출력핀의 방향을 설정

pinMode(핀번호, 방향) ;

- **pinMode(8, INPUT) ;**
- **pinMode(8, OUTOUT) ;**

디지털 입출력 API 실험



9번핀(PortB.1)을 입력핀으로
2번핀(PortD.2)을 출력핀으로
버튼이 눌렸을 때만 LED의 불이 들어 오도록 코드를 완성하시오.

```
void setup()
{
    pinMode(9, INPUT) ;
    pinMode(2, OUTPUT) ;
}

void loop()
{

}
```

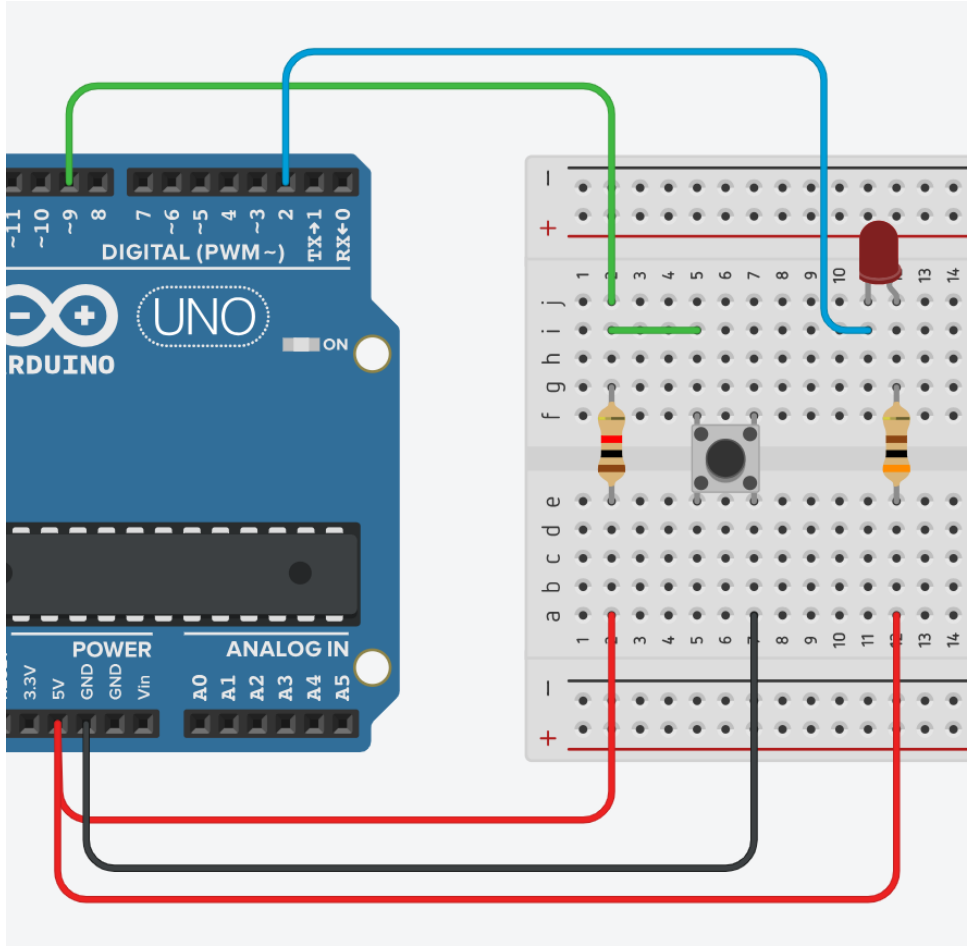
디지털 입출력 관련 API

- 디지털 입력

digitalRead(핀번호) ;

- **int input = digitalRead(8) ;**

디지털 입출력 API 실험



9번핀(PortB.1)을 입력핀으로
2번핀(PortD.2)을 출력핀으로
버튼이 눌렸을 때만 LED의 불이 들어 오도록 코드를 완성하시오.

```
void setup()
{
    pinMode(9, INPUT) ;
    pinMode(2, OUTPUT) ;
}

void loop()
{
    int input = digitalRead(9) ;
}
```

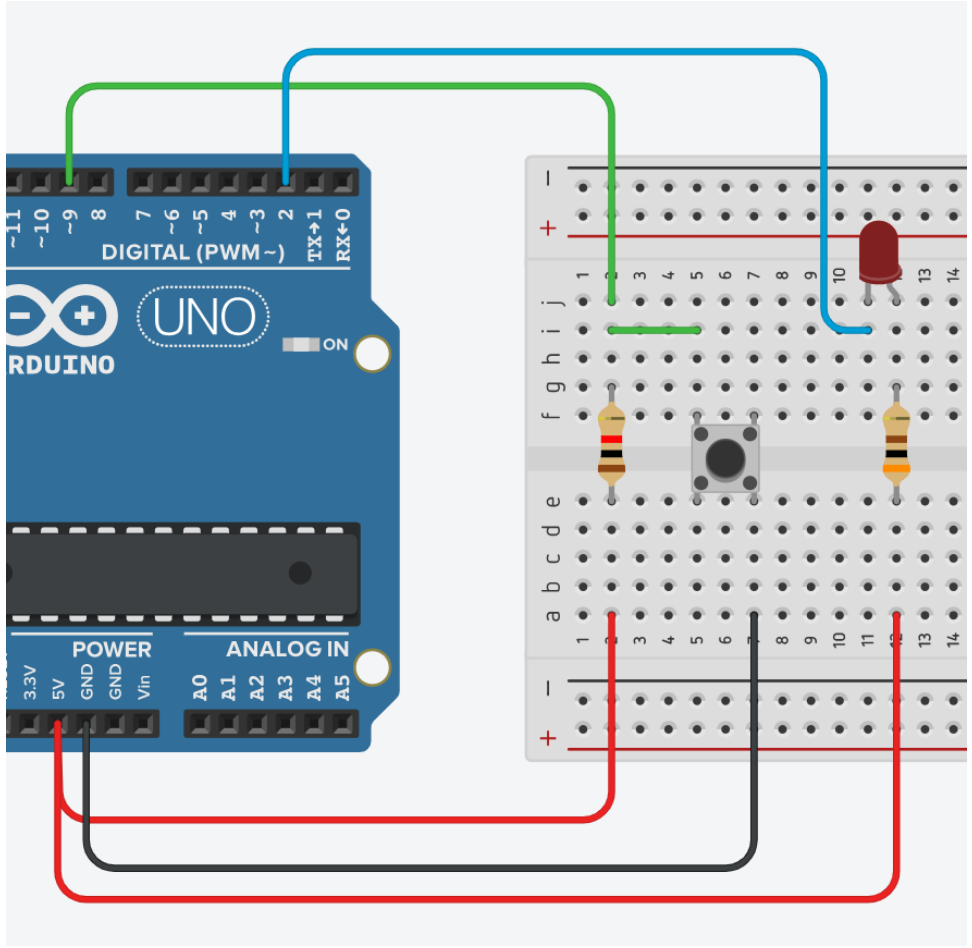
디지털 입출력 관련 API

- 디지털 출력

digitalWrite(핀번호, 출력레벨) ;

- **digitalWrite(8, 0) ;**
- **digitalWrite(8, 1) ;**
- **digitalWrite(8, LOW) ;**
- **digitalWrite(8, HIGH) ;**

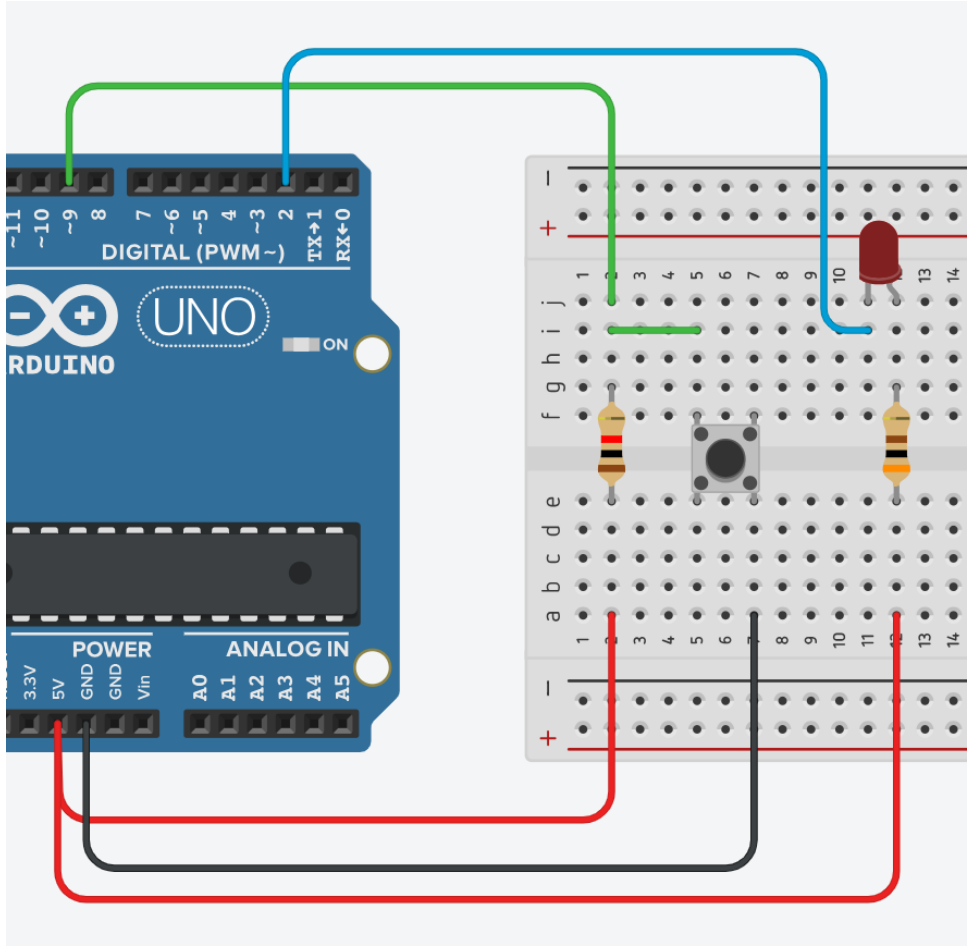
디지털 입출력 API 실험



9번핀(PortB.1)을 입력핀으로
2번핀(PortD.2)을 출력핀으로
버튼이 눌렸을 때만 LED의 불이 들어 오도록 코드를 완성하시오.

```
void setup()
{
    pinMode(9, INPUT) ;
    pinMode(2, OUTPUT) ;
}

void loop()
{
    int input = digitalRead(9) ;

    digitalWrite(2, LOW) ;
}
```

디지털 입출력 API 실험

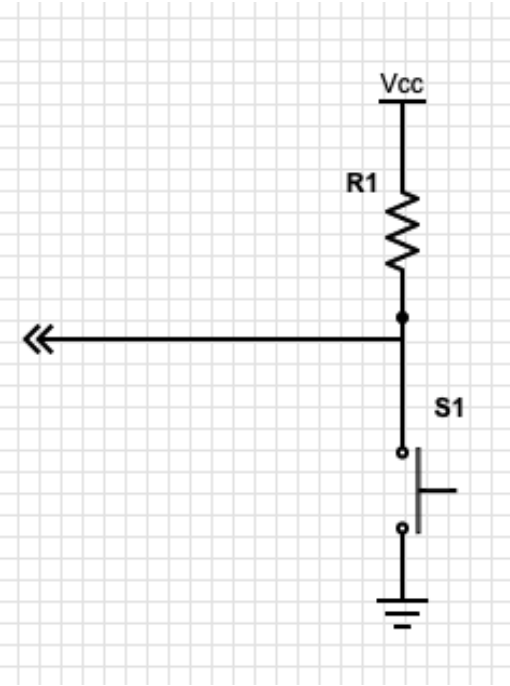


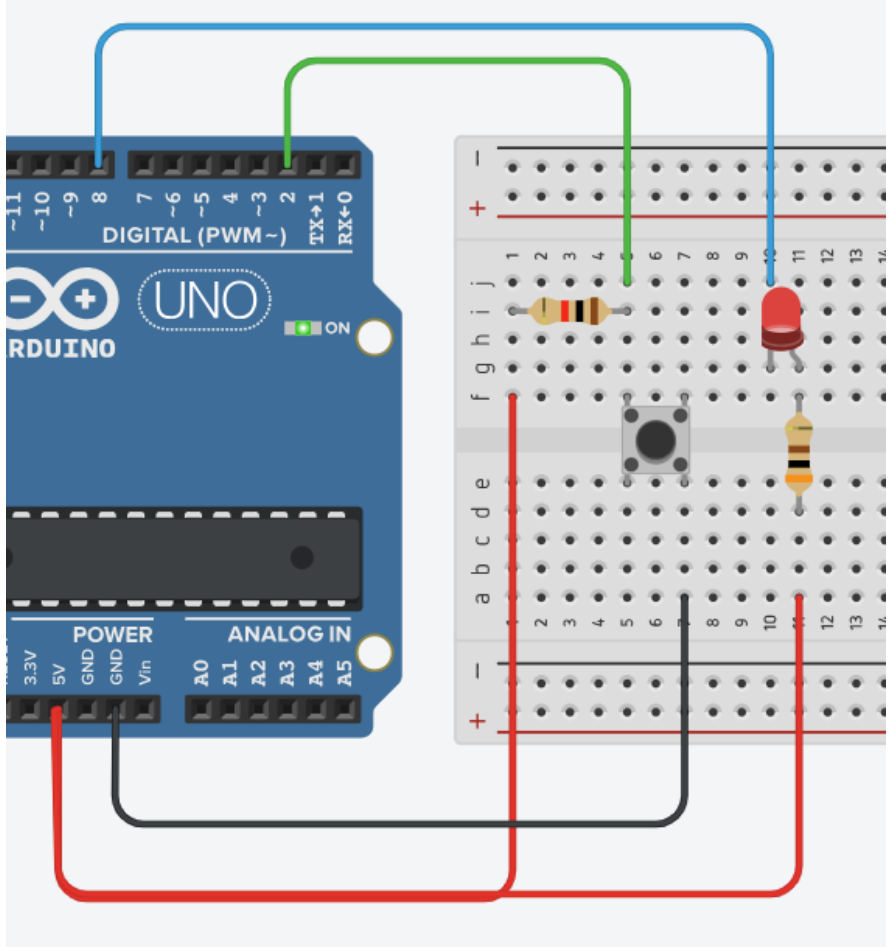
9번핀(PortB.1)을 입력핀으로
2번핀(PortD.2)을 출력핀으로
버튼이 눌렸을 때만 LED의 불이 들어 오도록 코드를 완성하시오.

```
void setup()
{
    pinMode(9, INPUT) ;
    pinMode(2, OUTPUT) ;
}

void loop()
{
    int input = digitalRead(9) ;

    if( input == 0 )
    {
        digitalWrite(2, LOW) ;
    }
    else
    {
        digitalWrite(2, HIGH) ;
    }
}
```

외부 인터럽트(External Interrupt)

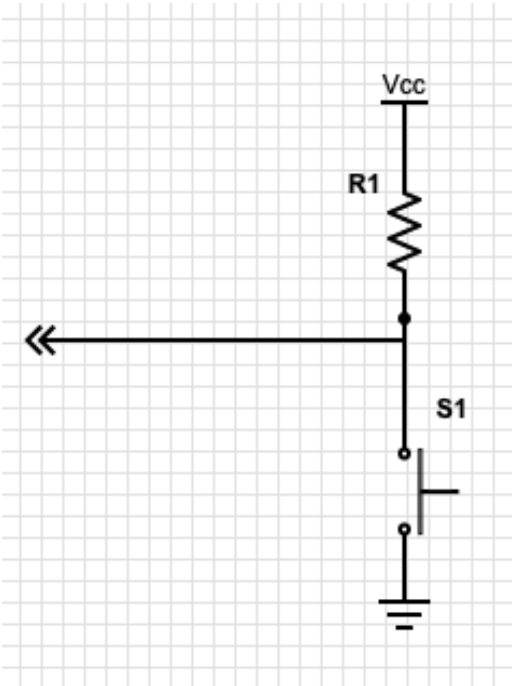
- **폴링** vs 인터럽트

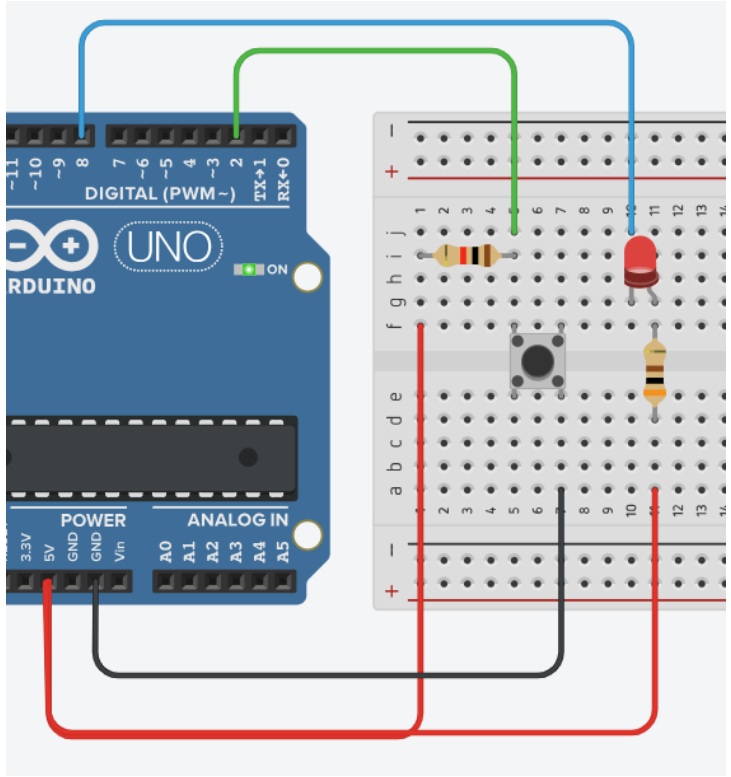




외부 인터럽트(External Interrupt)

- **폴링** vs 인터럽트

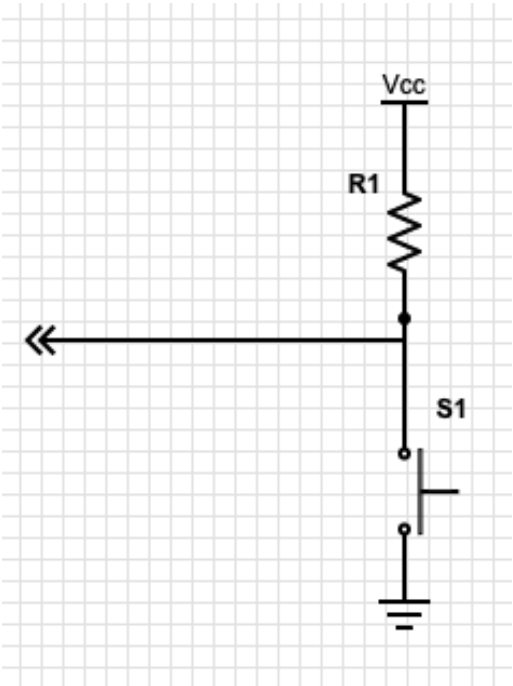


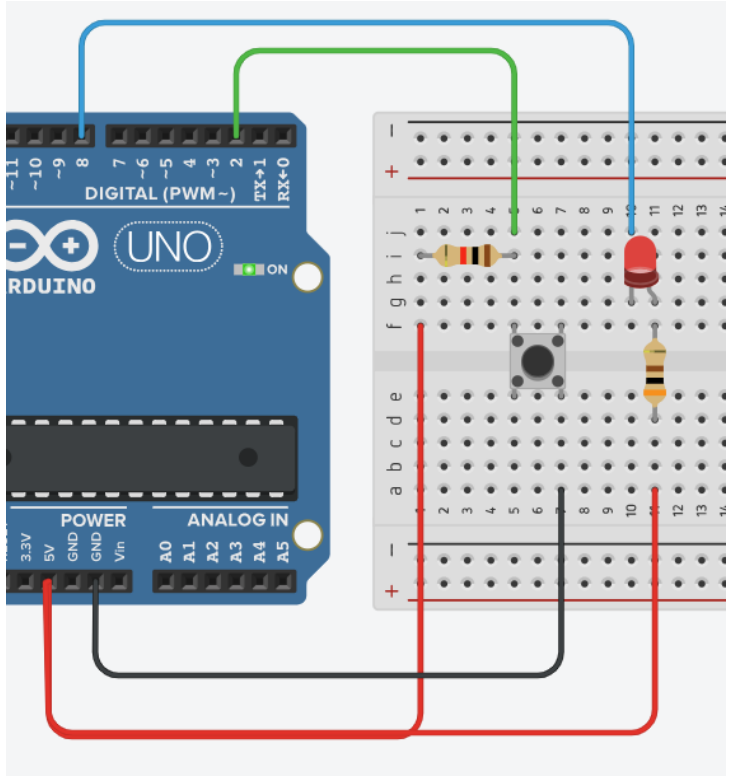



```
void setup()
{
  pinMode(2, INPUT) ;
  pinMode(8, OUTPUT) ;
}

void loop()
{
  int input = digitalRead(2) ;

  if( input == 0  )
  {
      digitalWrite(8, 0) ;
  }
  else
  {
       digitalWrite(8, 1) ;
  }
}
```

외부 인터럽트(External Interrupt)

- **폴링** vs 인터럽트

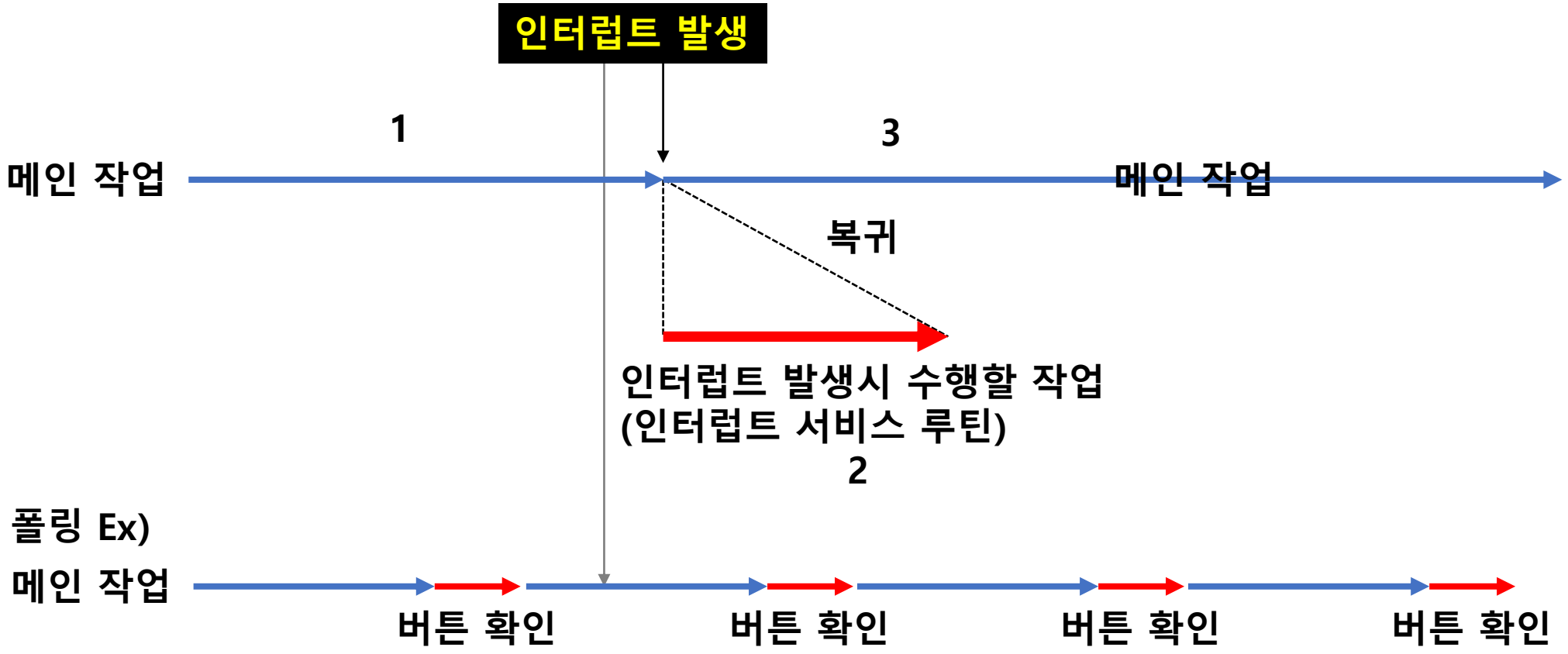





```
void setup()
{
  pinMode(2, INPUT) ;
  pinMode(8, OUTPUT) ;

  Serial.begin(9600) ;
}

void loop()
{

  digitalWrite(8, 0) ;
  delay(1000) ;

  digitalWrite(8, 1) ;
  delay(1000) ;

  int input = digitalRead(2) ;
  if( input == 0 )
  {
      Serial.println("key") ;
  }
}
```

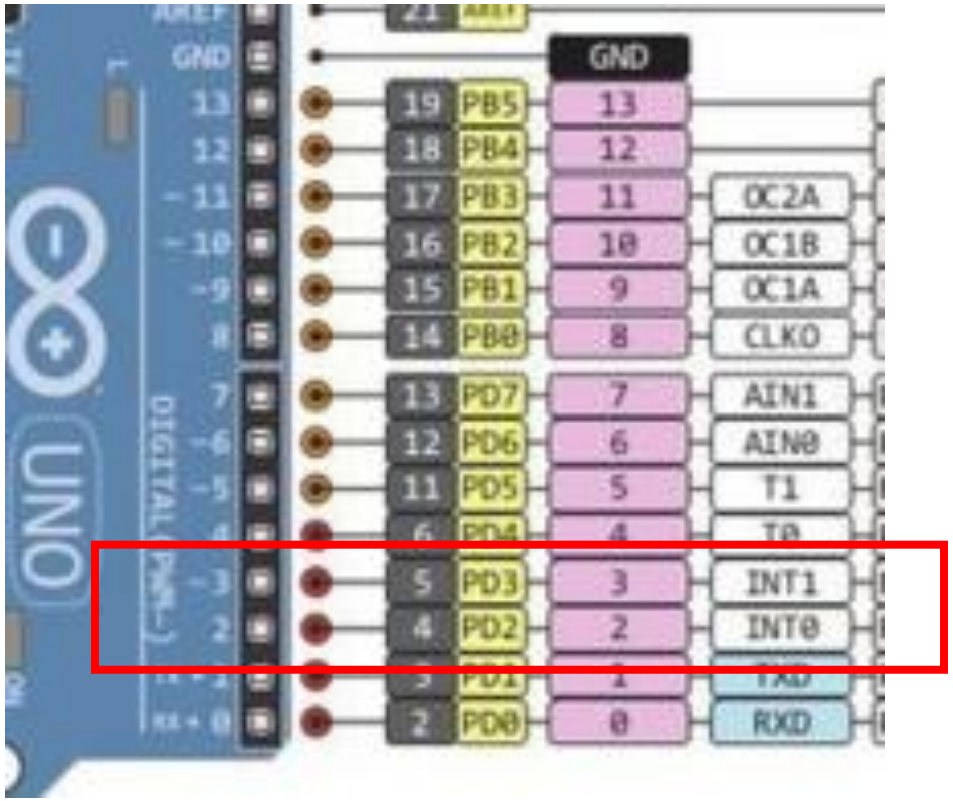
외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**



외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

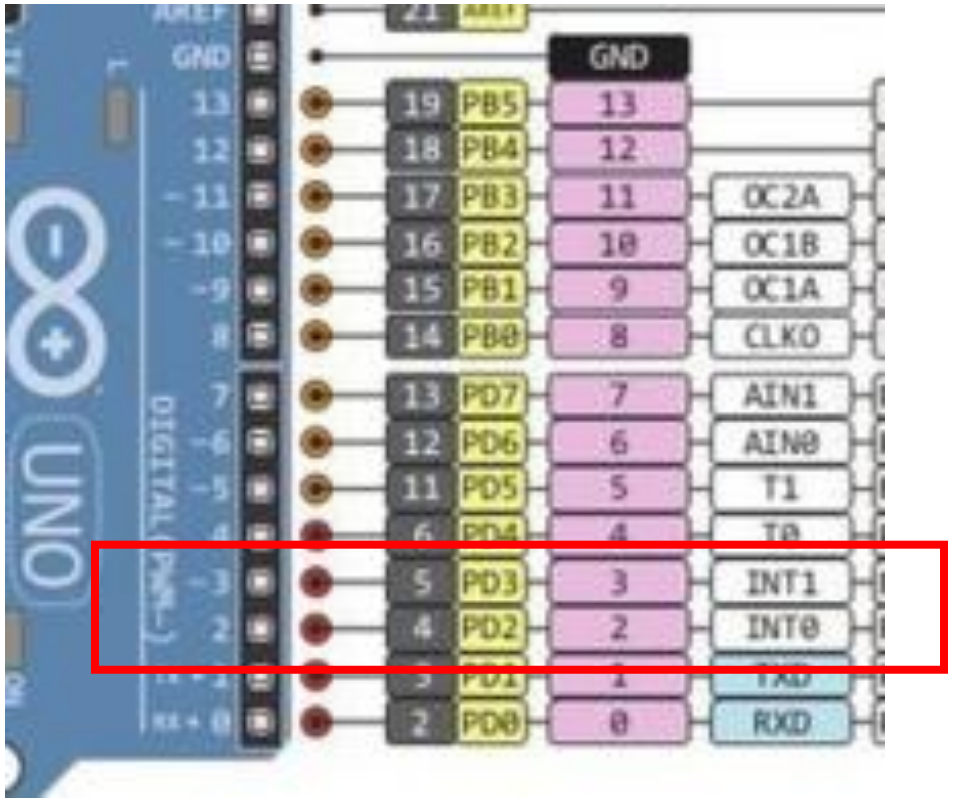


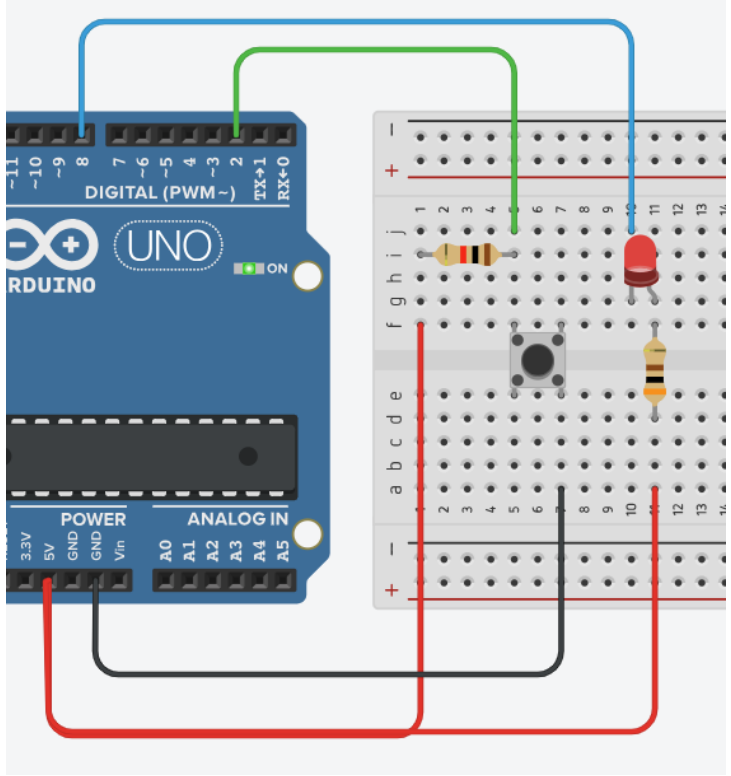
- **INT1 : Interrupt #1**
- **INT0 : Interrupt #0**

외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

2 → INT0 : Interrupt #0

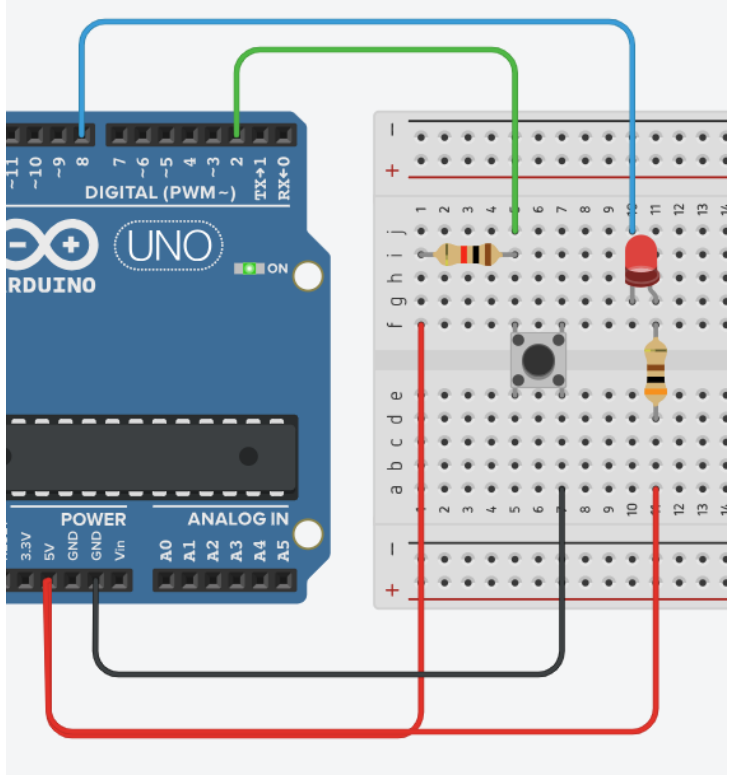




외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

2 → INT0 : Interrupt #0



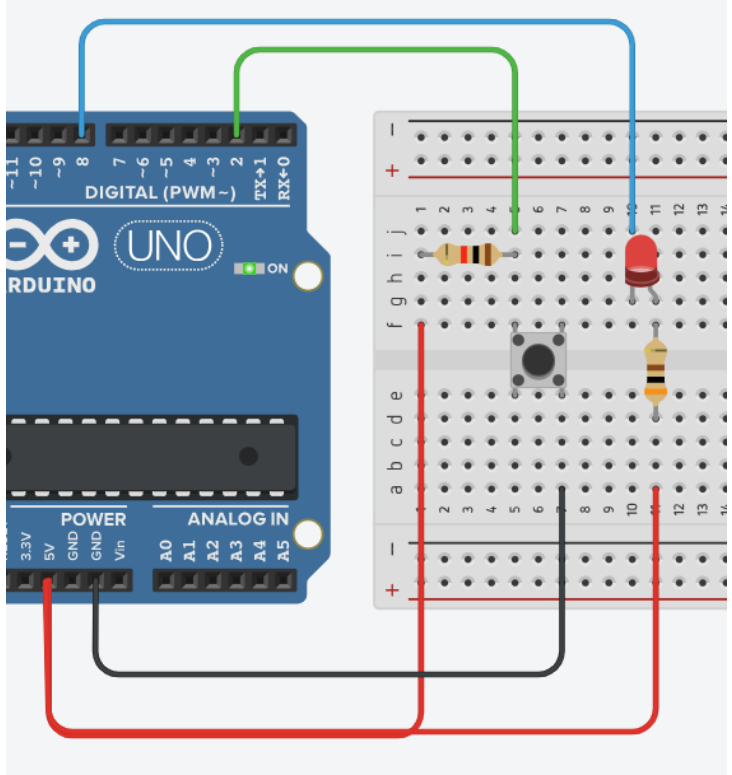
인터럽트 발동 조건 (mode)

모드	상태
LOW	**핀이 LOW일때**
CHANGE	**LOW->HIGH or HIGH->LOW로 변할 때**
RISING	**LOW ->HIGH일때**
FALLING	**HIGH -> LOW일때**
HIGH	**핀이 HIGH일때**

외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

2 → INT0 : Interrupt #0




```
attachInterrupt( digitalPinToInterrupt(핀번호), 서비스루틴함수명, 모드 );
```

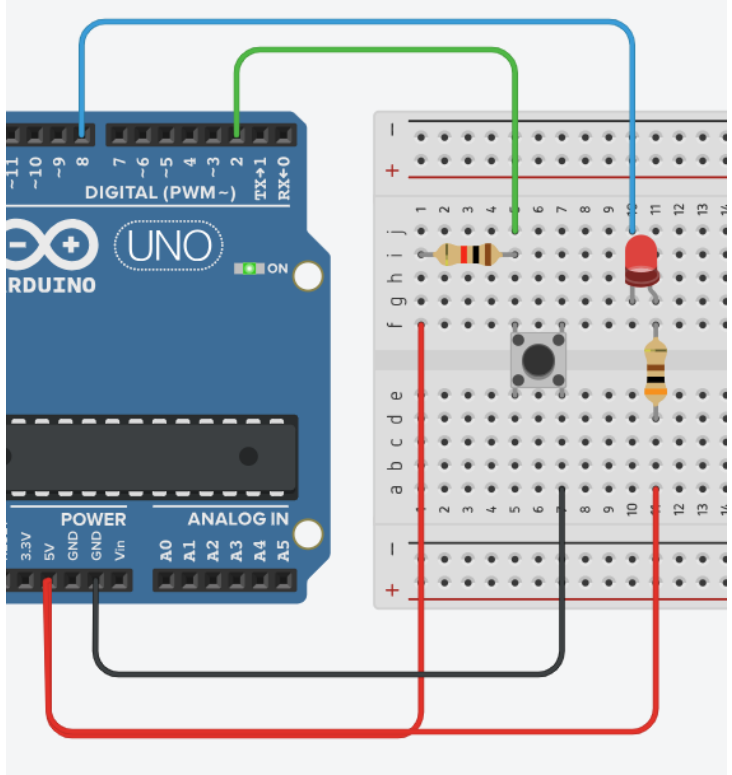
인터럽트 발동 조건 (mode)

모드	상태
LOW	**핀이 LOW일때**
CHANGE	**LOW->HIGH or HIGH->LOW로 변할 때**
RISING	**LOW ->HIGH일때**
FALLING	**HIGH -> LOW일때**
HIGH	**핀이 HIGH일때**

외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

2 → INT0 : Interrupt #0



attachInterrupt(digitalPinToInterrupt(2), ExINT, FALLING);

attachInterrupt(digitalPinToInterrupt(핀번호), 서비스루틴함수명, 모드);

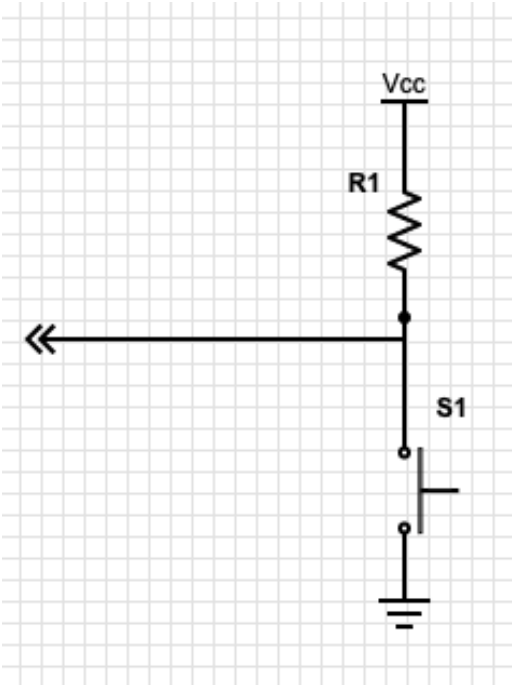
인터럽트 발동 조건 (mode)

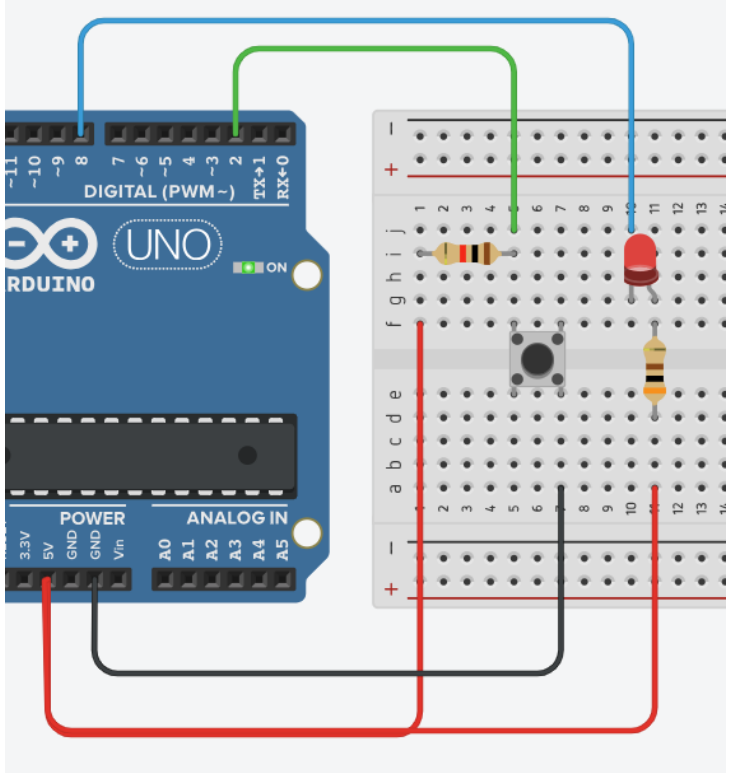
모드	상태
LOW	**핀이 LOW일때**
CHANGE	**LOW->HIGH or HIGH->LOW로 변할 때**
RISING	**LOW ->HIGH일때**
FALLING	**HIGH -> LOW일때**
HIGH	**핀이 HIGH일때**

외부 인터럽트(External Interrupt)

- 폴링 vs **인터럽트**

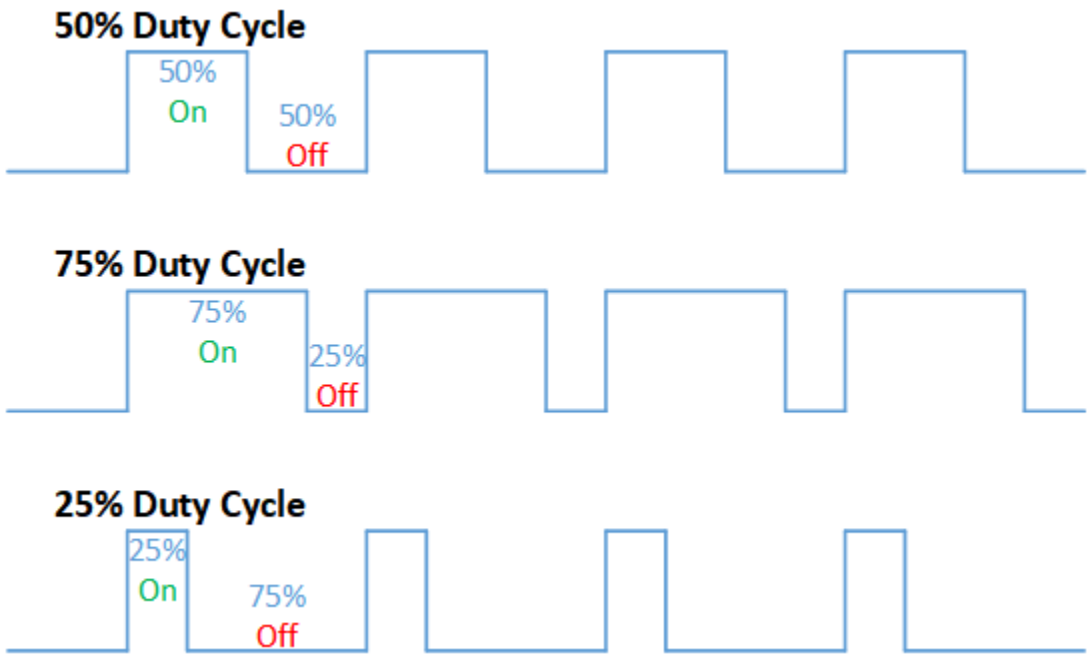
attachInterrupt(digitalPinToInterrupt(2), ExINT, FALLING);



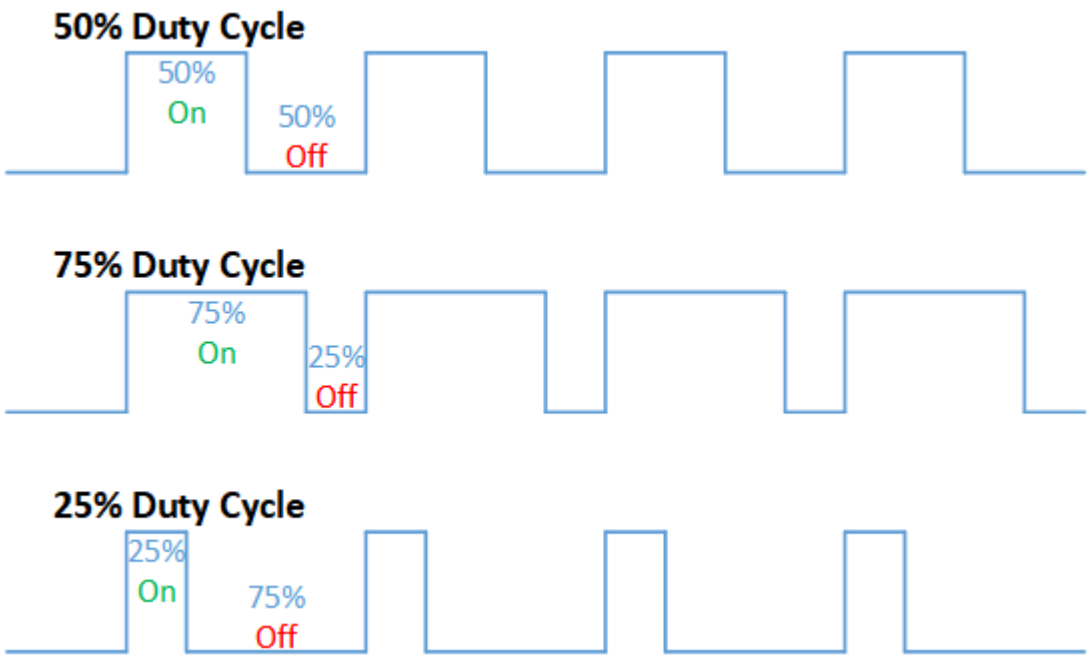


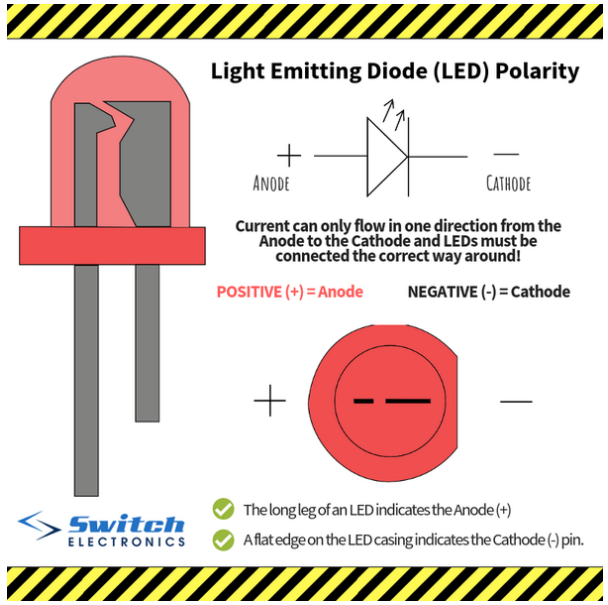

```
void setup()
{
  pinMode(8, INPUT) ;
  pinMode(2, OUTPUT) ;

  attachInterrupt( digitalPinToInterrupt(2), ExINT, FALLING );

  Serial.begin(9600) ;
}

void loop()
{
  digitalWrite(8, 0) ;
  delay(1000) ;

  digitalWrite(8, 1) ;
  delay(1000) ;
}

void ExINT()
{
  Serial.println("ExINT") ;
}
```

PWM(Pulse Width Modulation)



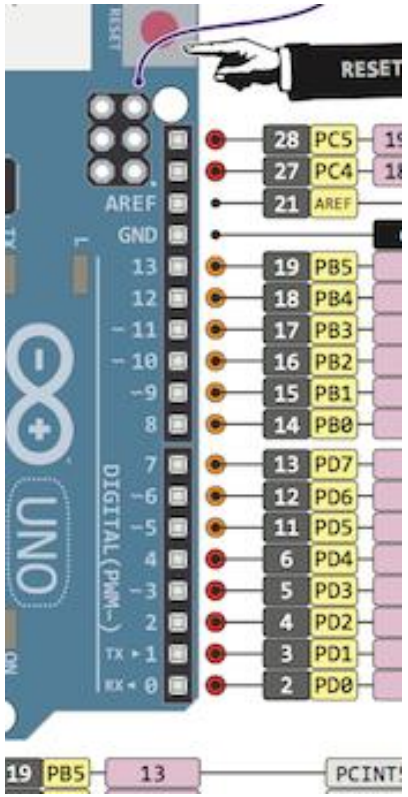
PWM을 이용한 LED 밝기 제어





디지털 입출력 관련 API

- PWM(디지털 출력) 관련 명령

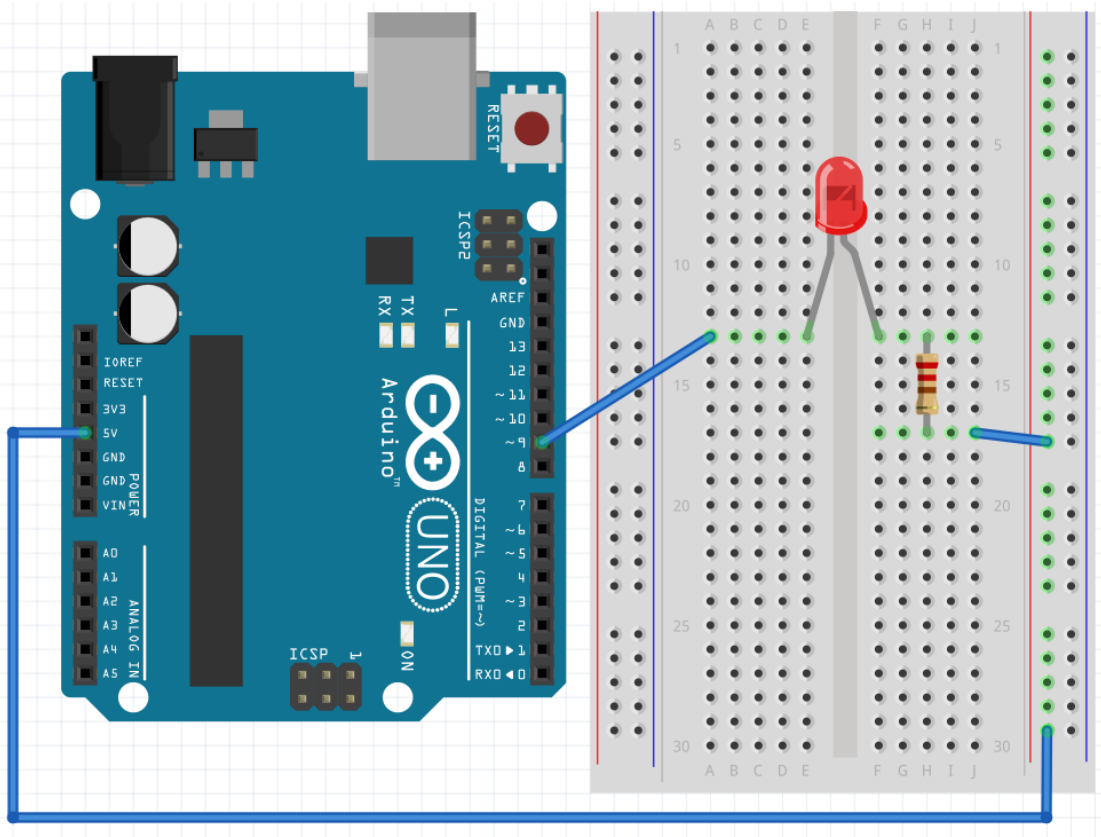


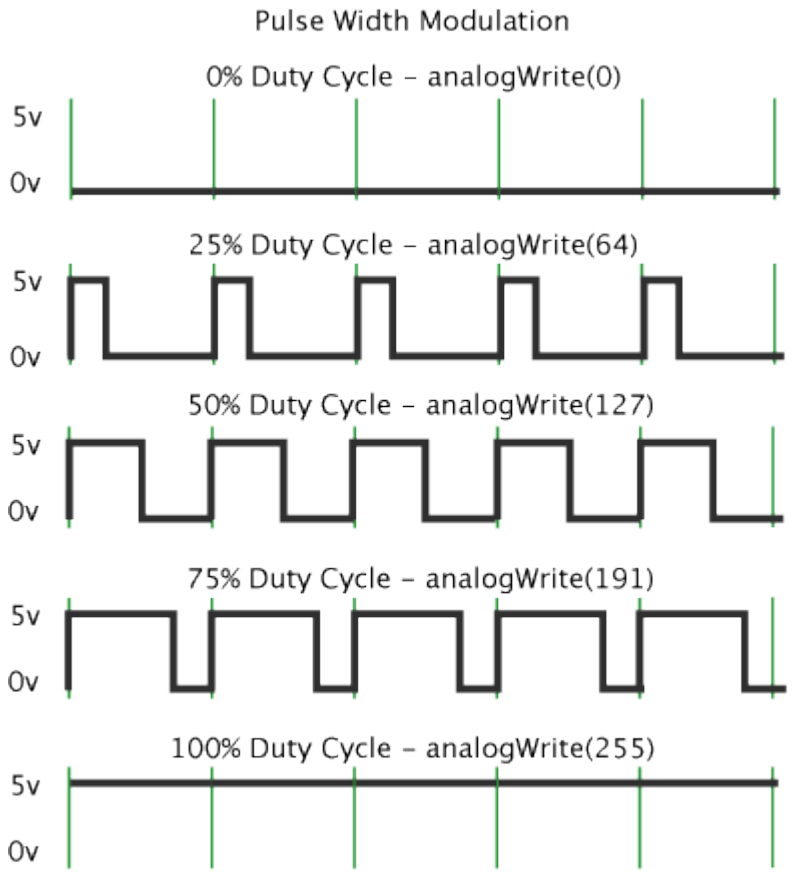
analogWrite(핀번호, Duty Cycle) ;

- **analogWrite(9, 0) ;**
- **analogWrite(9, 128) ;**
- **analogWrite(9, 255) ;**

아두이노를 이용한 LED 밝기제어 예제

- 함수 : analogWrite(핀번호, duty cycle)



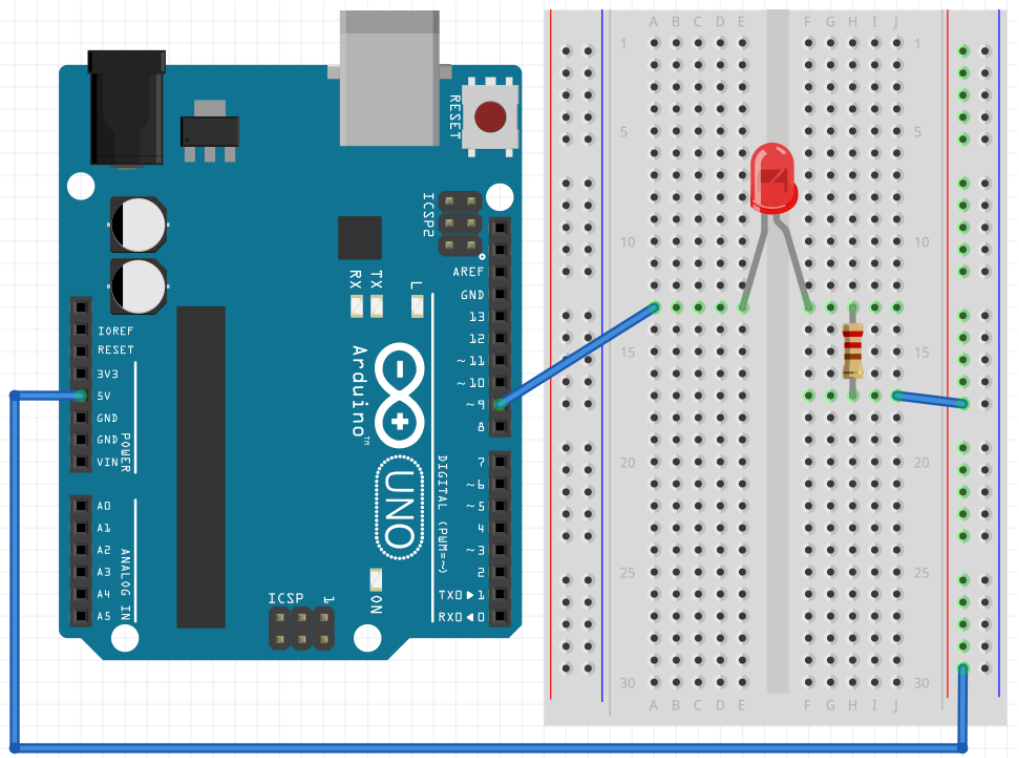


아두이노를 이용한 LED 밝기제어 예제

- 함수 : analogWrite(핀번호, duty cycle)

```
void setup() {
  pinMode(9, OUTPUT);  // 핀을 출력으로 설정
}

void loop() {
  analogWrite(9, 0); //analogWrite 값은 0 부터 255까지
}
```

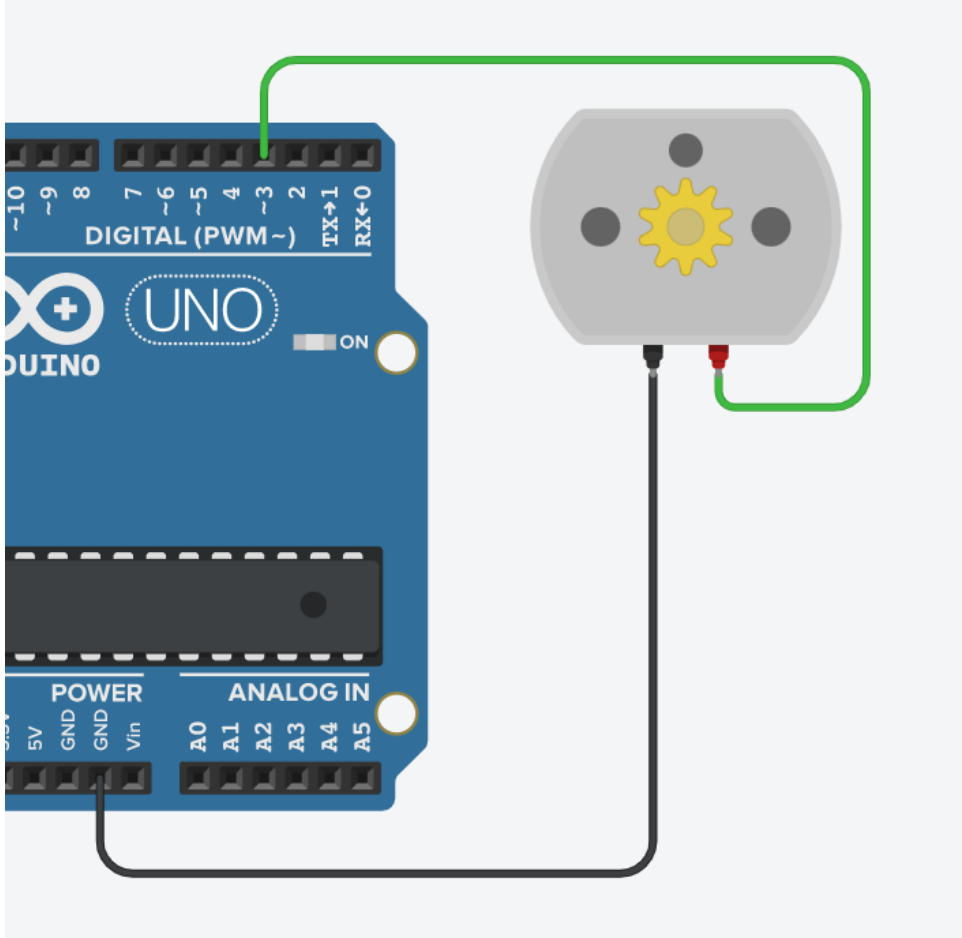
```
void setup() {
  pinMode(9, OUTPUT);  // 핀을 출력으로 설정
}

void loop() {
  analogWrite(9, 255); //analogWrite 값은 0 부터 255까지
}
```



analogWrite(PWM)을 이용한 DC모터 제어

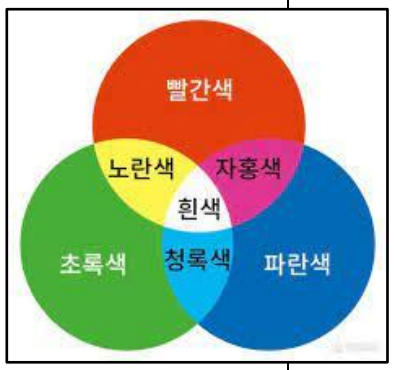
```
void setup()
{
  pinMode(3, OUTPUT);  // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, 255); //analogWrite 값은 0 부터 255까지
}
```

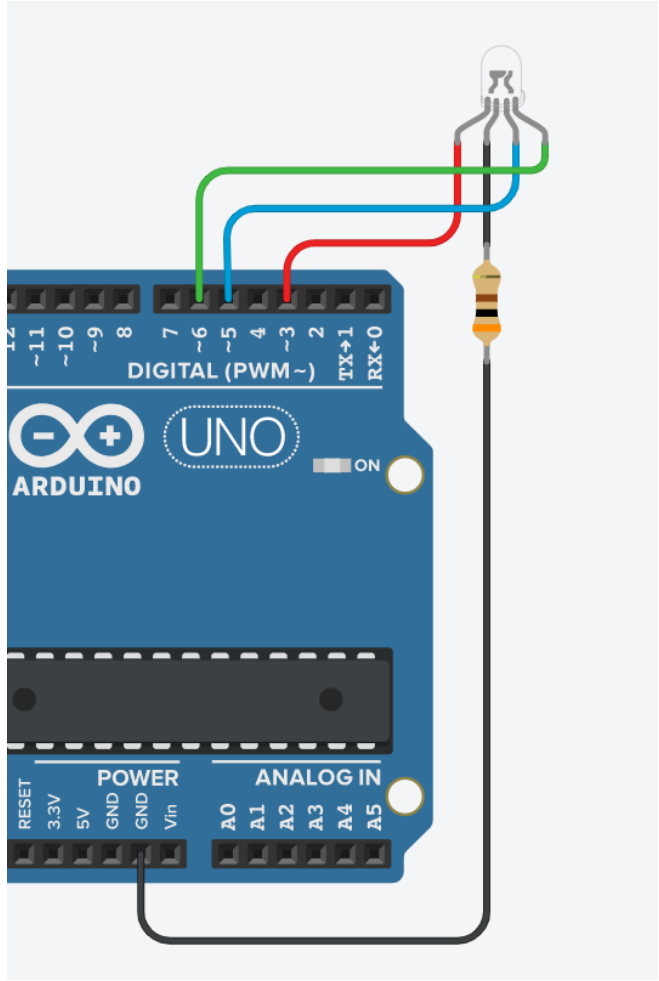
```
void setup()
{
  pinMode(3, OUTPUT);  // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, 128); //analogWrite 값은 0 부터 255까지
}
```



analogWrite(PWM)을 이용한 RGB LED제어

```
void setup()
{
  pinMode(3, OUTPUT);  // 핀을 출력으로 설정
  pinMode(5, OUTPUT);  // 핀을 출력으로 설정
  pinMode(6, OUTPUT);  // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, 255); //analogWrite 값은 0 부터 255까지
  analogWrite(5, 255); //analogWrite 값은 0 부터 255까지
  analogWrite(6, 0); //analogWrite 값은 0 부터 255까지
}
```

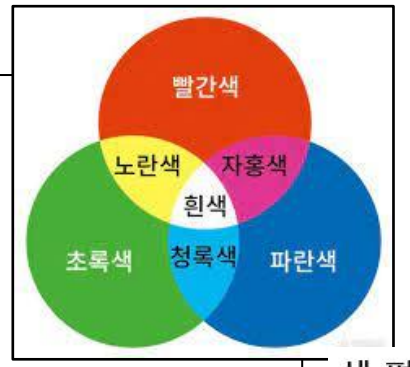


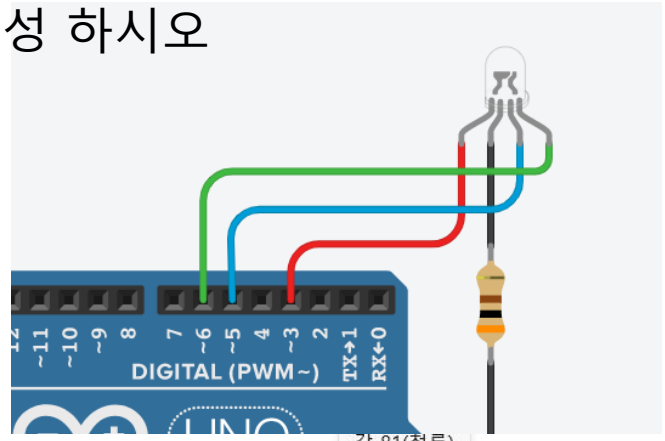


analogWrite(PWM)을 이용한 3색(full color) RGB LED제어

QUIZ : 3색 LED의 광량을 조절하여 아래의 **청록색**이 표현 되도록 코드를 완성 하시오

RGB(121, 207, 203)






```
void setup()
{
  pinMode(3, OUTPUT);  // 핀을 출력으로 설정
  pinMode(5, OUTPUT);  // 핀을 출력으로 설정
  pinMode(6, OUTPUT);  // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, [          ]); //analogWrite 값은 0 부터 255까지
  analogWrite(5, [          ]); //analogWrite 값은 0 부터 255까지
  analogWrite(6, [          ]); //analogWrite 값은 0 부터 255까지
}
```